NETWORK TESTING PRACTICAL

1.Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.

Ans.

Procedure:

step.1 First, connect my laptop and my friend's mobile phone to the same Wi-Fi network.

step.2 On the mobile phone, open Wi-Fi settings and check its IP address.

step.3 On my laptop, open Command Prompt by typing cmd in the Windows search bar.

step.4 Type ipconfig and press Enter to check the laptop's IP address.

step.5 Now type the following command in Command Prompt, replacing the example IP with my friend's phone IP address.

ping 192.168.1.5

step.6 Press Enter and wait for the result.

step.7 If the reply messages appear, the devices can communicate with each other. If the request times out, communication may be blocked by the phone's settings or the Wi-Fi network.

Example Output:

Pinging 192.168.1.5 with 32 bytes of data:

Reply from 192.168.1.5: bytes=32 time=15ms TTL=64

Reply from 192.168.1.5: bytes=32 time=12ms TTL=64

Reply from 192.168.1.5: bytes=32 time=14ms TTL=64

Reply from 192.168.1.5: bytes=32 time=13ms TTL=64

2.Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

Ans.

Procedure:

step.1 First, connect my laptop to the internet.

step.2 Open Command Prompt by typing cmd in the Windows search bar.

step.3 Type the following command and press Enter.

tracert www.youtube.com

step.4 The command shows the route taken by data packets from my laptop to the destination server.

step.5 Each numbered line represents a hop, usually a router or network device through which the packets travel.

step.6 Wait until the command finishes and note down the total number of hops displayed.

Example Output:

over a maximum of 30 hops:

1	2 ms	1 ms	2 ms	192.168.1.1
2	10 ms	12 ms	11 ms	ISP Gateway
3	18 ms	16 ms	19 ms	Network Router
4	25 ms	23 ms	26 ms	Destination Network

3.Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

Ans.

Procedure:

step.1 First, connect my laptop and my friend's mobile phone to the same Wi-Fi network.

step.2 Open Command Prompt on my laptop and type ipconfig to check the network details.

step.3 Note down my friend's mobile phone IP address.

step.4 Disconnect my friend's mobile phone from the Wi-Fi network.

step.5 Now, open Command Prompt on my laptop and enter the following command:

ping 192.168.1.5

step.6 Press Enter and observe the output displayed on the screen.

Example Output:

Pinging 192.168.1.5 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

Observation:

- After disconnecting the mobile phone from Wi-Fi, the laptop does not receive a reply from the phone.
- The message "Request timed out" appears on the screen.

- All four packets are lost in this example because the phone is disconnected or not responding.

4.Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

Ans.

Procedure:

step.1 First, I connected my laptop to the home Wi-Fi network.

step.2 I opened Command Prompt by typing cmd in the Windows search bar.

step.3 I entered the command arp -a and pressed Enter.

step.4 The command displayed a list of IP addresses and physical (MAC) addresses of devices that my laptop had recently communicated with on the local network.

step.5 I identified the devices using their IP addresses and checked which person or gadget they belonged to.

Table: IP Addresses of Connected Devices

Sr. No.	Device Name	IP Address	Belongs To
1	Wi-Fi Router	192.168.1.1	Home Network
2	Laptop	192.168.1.5	Me
3	Smartphone	192.168.1.6	My Father
4	Smartphone	192.168.1.7	My Mother
5	Smart TV	192.168.1.8	Living Room
6	Smartphone	192.168.1.9	My Brother

5.Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Ans.

1. Introduction

The main purpose of this practical was to set up a network using a home Wi-Fi connection and check whether the connected devices could communicate with each other and access the internet. I used my laptop and a smartphone to perform the connectivity tests.

2. Network Setup Procedure

First, I connected my laptop and a smartphone to the same Wi-Fi network. I checked that both devices were connected properly and that the Wi-Fi connection was working. After that, I opened Command Prompt on my laptop to perform the network tests.

3. Testing Connectivity Using Ping

I used the ping command to check the connection between my laptop and the other device. I entered the IP address of the device in Command Prompt.

For example, I used the command ping 192.168.1.6.

The ping command sent data packets to the selected device and checked whether it responded. When replies were received, it showed that the devices could communicate over the network. If the request timed out, it could mean that the device was disconnected or that its firewall was blocking ping requests.

4. Testing the Route Using Traceroute

Next, I used the tracert www.youtube.com command in Command Prompt. This command showed the network hops through which the data travelled towards the destination. It helped me understand how data passes through different routers before reaching a website.

Some hops responded with their IP addresses and response times, while a few could show a request timeout because some routers do not respond to traceroute requests.

5. Observations and Issues

During the practical, I learned that the ping command is useful for checking network connectivity, while tracert helps identify the route taken by data packets. I also learned that a timeout does not always mean the internet connection is faulty. It may happen because of firewall settings, network restrictions, or routers that do not respond to these requests.

I also learned that the arp -a command can display IP and MAC address information for devices recently discovered through local network communication. However, it may not show every device connected to the Wi-Fi.

6. Conclusion

This practical helped me understand how to set up a basic Wi-Fi network and test connectivity using Command Prompt. I learned how to use ping, tracert, and arp -a for basic network troubleshooting. These commands are useful for identifying connectivity problems and understanding how devices communicate over a network.